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To cite this article: Yang Pan , Guangfeng Wu , Huifang Ma , Shujie Zhou & Huixuan Zhang (2020): Improved compatibility of PET/HDPE blend by using GMA grafted thermoplastic elastomer, Polymer-Plastics Technology and Materials, DOI: [10.1080/25740881.2020.1765382](https://doi.org/10.1080/25740881.2020.1765382)

To link to this article: <https://doi.org/10.1080/25740881.2020.1765382>



Published online: 27 Jun 2020.



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Improved compatibility of PET/HDPE blend by using GMA grafted thermoplastic elastomer

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ABSTRACT

Herein, graft-modified ethylene-1-octene copolymer (POE-*g*-GMA) and styrene-butadiene-styrene triblock copolymer (SBS-*g*-GMA) were found to be excellent reactive compatibilizers for immiscible poly(ethylene terephthalate) (PET)/high-density polyethylene (HDPE) blends via *in-situ* reaction compatibilization. With increase in compatibilizer amount, uniform phase morphology was observed in all the blends. Thus, exhibiting enhanced mechanical properties, especially, the notched Izod impact strength. In comparison with SBS-*g*-GMA, compatibilizer POE-*g*-GMA demonstrated greater impact on the compatibility. The addition of 15% POE-*g*-GMA produced blends with best mechanical properties. Besides, both POE-*g*-GMA and SBS-*g*-GMA enhanced the melt viscosity of PET/HDPE blends.

ARTICLE HISTORY

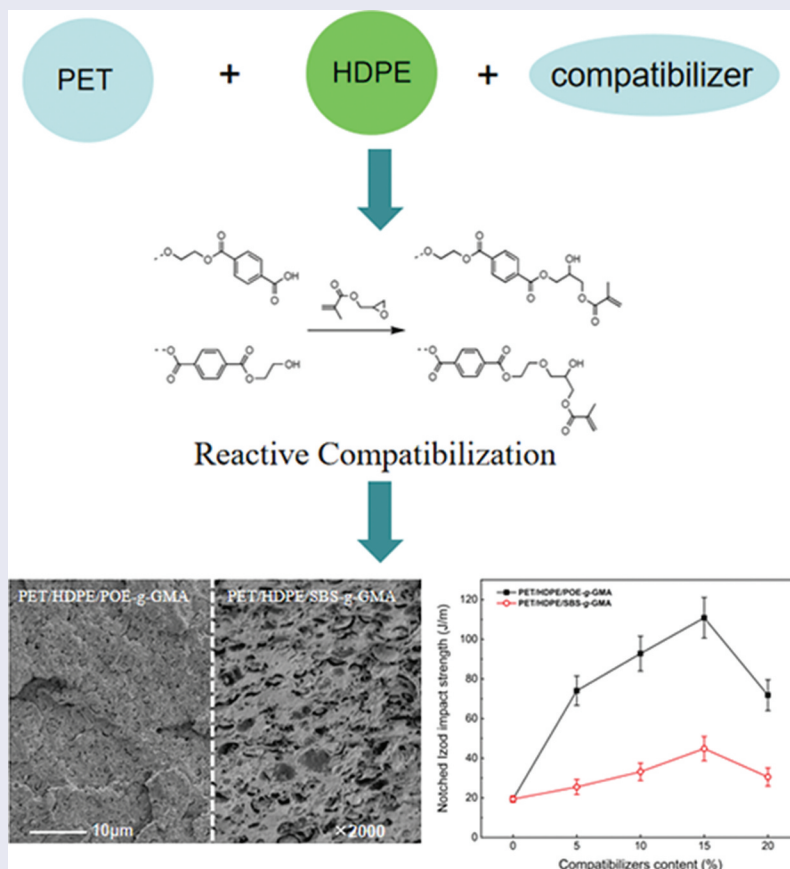
Received 18 February 2020

Revised 2 May 2020

Accepted 3 May 2020

KEYWORDS

Compatibilizer; poly(ethylene terephthalate); thermoplastic elastomer; high-density polyethylene; glycidyl methacrylate



1. Introduction

The compatibilization of immiscible polymer blends is an attractive technique to obtain high-performance

materials with properties that are generally not attainable in either single pure component.^[1,2] The addition of block or graft copolymers that are made up of two components in immiscible blends and reactive

compatibilization are two primary methods. For reactive compatibilization, a functionalized polymeric additive is generally necessary which can undergo a chemical transformation with one of the blend components, yielding in situ copolymers at the interfaces of the two immiscible blends. Thus, reactive compatibilization is significantly more efficient than traditional block or graft copolymers, as it does not rely on the diffusion of pre-formed copolymers and the functional groups are either already present or easily grafted on the blend components.

Poly(ethylene terephthalate) (PET) and polyethylene (PE) are two of the most common thermoplastic resins. PET has excellent heat resistance, hardness, and chemical stability, while polyethylene has excellent toughness. Blending of PET and PE is particularly advantageous as novel materials with promising property combinations can be obtained at relatively low cost.^[3–6] Several studies on the reactive compatibilization of PET/PE blends have been performed to improve their miscibility. Pietrasanta et al.^[7] reported that glycidyl methacrylate functionalized polyolefins are effective compatibilizers for PET/PE system, as the epoxy group in the compatibilizers can react with hydroxyl and carboxyl of PET. The mechanical properties of blends were greatly improved. Also, ethylene/glycidyl methacrylate copolymer (E/GMA) is suggested to be effective in compatibilizing HDPE/PET blends via *in-situ* formation of compatibilizer.^[8] Zhang et al.^[9] employed SEBS-*g*-MA as a compatibilizer for recycled-PET/PE blends. The group found that the compatibilizer was located between the two-phase interface and the morphology and mechanical properties of the blends depended to a large extent on the miscibility between SEBS-*g*-MA and polyethylene.

In general, reactive functional agents grafted elastomers are commonly used for reactive compatibilizing polyester/polyolefin blends. The most widely used elastomers include SEBS, ethylene-propylene copolymer (EPR), and ethylene propylene diene monomer (EPDM),^[10–12] and the most common reactive monomers are acrylic acid (AA), maleic anhydride (MA), and epoxy groups.^[13–17] Ethylene-1-octene copolymer (POE) is a novel thermoplastic elastomer which exhibits multiple excellent properties. The modification of POE has become one of the current research hotspots.^[18–22] Styrene-butadiene-styrene triblock copolymer (SBS) is one of the most common thermoplastic elastomers. SBS and modified SBS have been widely used in toughening nylon and modifying asphalt.^[23–28] However, there are few reports on the use of functionalized POE or SBS as compatibilizers for PET/HDPE blends.

Inspired by the outstanding properties of POE and widely used SBS, and in efforts to expand the scope of

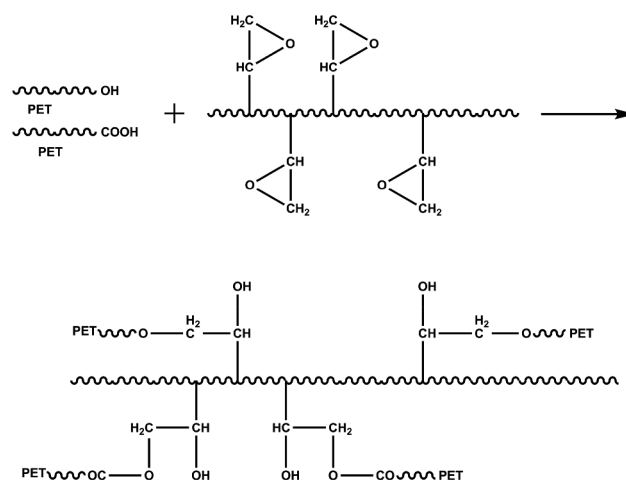


Figure 1. Schematic of compatibilization reaction mechanism.

reactive compatibilization for PET/HDPE blends, herein, POE-*g*-GMA and SBS-*g*-GMA were used to compatibilize PET/HDPE blends, respectively. POE-*g*-GMA and SBS-*g*-GMA were synthesized by grafting glycidyl methacrylate (GMA) onto POE and SBS, respectively. There were two reasons for choosing GMA instead of maleic anhydride (MA) in the present experiment. Firstly, MA can only react with hydroxyl end groups, while GMA can both react with hydroxyl and carboxyl end groups of PET. The schematic of compatibilization reaction mechanism is shown in Figure 1. Also, the esterification reaction is reversible under high-temperature conditions.^[29] Thereafter, PET/HDPE/POE-*g*-GMA and PET/HDPE/SBS-*g*-GMA ternary blends were prepared using a co-rotating twin screw extruder and injection molding machine. The amount of POE-*g*-GMA and SBS-*g*-GMA were changed from 0% to 20%, based on the whole weight of PET and HDPE binary system. The effect of these two synthesized compatibilizers on the morphology, mechanical properties, and thermal properties of PET/HDPE (80:20 w/w) blends were investigated in detail.

2. Experimental

2.1. Materials

The pellet form PET (56151) with an intrinsic viscosity of 0.81 ± 0.02 dl/g, $T_m = 248 \pm 1^\circ\text{C}$ was obtained from DuPont. HDPE (7210) was bought from LG Chem in pellet. (MFR 5.5 g/10 min; 190°C , 2.16 kg). POE (ENGAGE 8150) was bought from The Dow Chemical Company. SBS (F675) was supplied by SINOPEC Maoming Company with $40 \pm 2\%$ styrene content. GMA (97% purity), styrene (St, 99% purity), and dicumyl peroxide (98% purity) were purchased from Aladdin Biochemical Co. Ltd. (Shanghai, China) and

were analytical grade. Paraffin liquid (chemical purity) was from Fuchen (TianJin) Chemical Reagents Co. Ltd.

2.2. Grafting reactions

POE-g-GMA and SBS-g-GMA were formulated by melting grafting of POE and SBS in a co-rotating twin screw extruder (HT-30, L/D = 35, Nanjing Rubber & Plastics Machinery Plant Co. Ltd., China), with GMA as the grafting monomer and St as the grafting co-monomer. The role of St was to increase the grafting degree of GMA.^[30] The eight barrel temperatures from the feeding to die zones were set as 120, 160, 165, 170, 175, 180, 185, and 180°C, respectively. The screw speed was 50 rpm. The composition and gel content of SBS-g-GMA are listed in Table 1. The minimum gel amount of SBS-g-GMA occurred when the amount of St was 24 g, and a smaller gel content was beneficial to increase the compatibilizing efficiency of the compatibilizer. To ensure consistency, the fixed dosages of GMA and St in POE-g-GMA were 56 g and 24 g, respectively. Regardless of the amount of GMA and St, the gel content of POE-g-GMA was extremely low for accurate measurements. The composition of the grafted polymers is presented in Table 2. The grafted product of SBS (2.5 g W_1) was dissolved in carbon tetrachloride at 65°C for 60 minutes. Then, the solution was filtered under constant stirring, and finally, unreacted DCP, GMA, and gel were removed by precipitating the filtrate into excess ethanol. The gel was then dried for 2 hours in a drying oven at 60°C and weighed and recorded as W_2 . The gel content was calculated according to the following formula,

$$\text{Gel content} = \frac{W_2}{W_1} \times 100\% \quad (1)$$

Similarly, the grafted product of POE was heated to reflux in xylene for 90 minutes, and then, the solution was precipitated with excess acetone. The precipitate was then cleaned 2–3 times with appropriate amount of acetone, and the purified grafted POE was obtained.

Table 1. Composition and gel content of SBS grafting copolymer.

SBS (g)	GMA (g)	St (g)	Gel content (%)
1000	14	6	76.5
1000	28	12	55.3
1000	42	18	29.3
1000	56	24	0.08
1000	70	30	33.4

Table 2. Composition of grafted polymers.

POE (g)	SBS (g)	GMA (g)	St (g)	DCP (g)	Paraffin liquid (g)
1000	-	56	24	4	0
-	1000	56	24	4	100

The two purified grafted products were dried in a vacuum oven for further use.

2.3. Blends preparation

Before the blending process, PET pellets were dried in a vacuum oven for 12 h at 120°C, and the compatibilizers were dried at 60°C for 3 h. PET/HDPE (80:20 w/w) binary blends, PET/HDPE/POE-g-GMA, and PET/HDPE/SBS-g-GMA ternary blends were prepared using a co-rotating twin-screw extruder. In the ternary blends, the amount of compatibilizer were 5%, 10%, 15%, and 20%, respectively. Extrusion temperature for the extruder from zone 1 to zone 8 were 180°C, 220°C, 230°C, 240°C, 245°C, 250°C, 255°C, and 250°C, respectively. The die temperature was 250°C while the screw speed was 80 rpm. The extrudates were cooled in a water sink, pelletized, and dried in a vacuum oven for 6 h at 80°C. Subsequently, the dried pellets were injection molded using an injection molding machine (MA860/260 G Haitian Plastics Machinery Co. Ltd., China) at 240–255°C. Impact specimens with the size of 63.5 mm×12.7 mm×6.2 mm and dumbbell-shaped tensile specimens with effective size of 98 mm×13 mm×3.2 mm were obtained. All the test specimens were placed at room temperature for 48 hours.

2.4. Analysis and characterization

2.4.1. FTIR spectroscopy

FT-IR spectra of grafted POE, grafted SBS, pure POE, and pure SBS were monitored using a Nicolet 460 FT-IR spectrometer with ATR (Attenuated Total Reflection) accessory in the range of 4000–650 cm^{-1} . This analysis was performed to confirm whether GMA monomers were successfully grafted onto POE or SBS.

2.4.2. Determination of grafting degree of GMA

The grafting degree (G_d) of GMA was determined using a back-titration method. Purified POE-g-GMA (0.5 g) was dissolved in 50 ml of xylene, the mixture was then heated to reflux for 1 h at 110°C. Then, 10 ml of isopropanol solution of HCl (0.1 mol/L) was added to the solution and the mixture solution was heated to reflux for 100 min. Thereafter, it was back-titrated with ethanol solution of KOH (0.1 mol/L) and the bromophenol blue was used as an indicator. The grafting degree of SBS-g-GMA was also obtained by the above method. The grafting degree of GMA was calculated using the following equation:

$$G_d = \frac{(V_0 - V_1) \times C \times 142.15}{W \times 1000} \times 100\% \quad (2)$$

where V_0 is the volume of ethanol solution of KOH consumed by pure POE or pure SBS as a reference (ml), V_1 is

the volume of ethanol solution of KOH consumed by grafted POE or SBS (ml), C is the molar concentration of isopropanol solution of HCl (mol/L), and W is the weight of the samples (g).

2.4.3. Morphological observations

The morphology of the blends was investigated using a scanning electron microscope (SEM JSM 5600, Japan). The samples were cryogenically fractured in liquid nitrogen and etched with xylene for 48 h to remove the POE-g-GMA and SBS-g-GMA phase from the specimens. Then, the etched samples were dried and coated with a thin gold layer before SEM analysis. The statistical average diameter of the dispersed HDPE phase was calculated using ImageJ software which analyzed about 150 particles.

2.4.4. Differential scanning calorimetry (DSC)

All the samples were analyzed using a Perkin Elmer Pyris DSC differential scanning calorimeter (DSC). First, the samples (5–8 mg) were heated from 30°C to 280°C, and then, held for 5 min to remove prior thermal history. Thereafter, the samples were cooled to 40°C at a cooling rate of 10°C/min and were heated again to 280°C at 10°C/min.

The crystallinity of the PET and HDPE components was determined using the following relationship,

$$X_d = \frac{\Delta H_m}{\phi \times \Delta H_m^0} \times 100 \quad (3)$$

where ΔH_m is the measured melt enthalpy for PET or HDPE of blends, ΔH_m^0 is the assumed heat of crystallization of completely crystalline PET (140 J/g) or HDPE (293 J/g), and ϕ is the weight fraction of PET or HDPE in the blends.

2.4.5. Mechanical properties

The specimens for the mechanical property testing were obtained from the injection molding process. The tensile properties were tested using INSTRON 3365 Mechanical Properties Testing Machine (Instron Corporation) at the crosshead speed of 50 mm/min at room temperature, according to the Standard of China GB/T1040-92. A notch of depth of 2.55 mm, angle of 45°, and notch radius of 0.25 mm was milled. Notched Izod impact strength was performed using XJU-22 (Chengde, China) Izod impact test machine at 23°C, according to the Standard of China GB1843-80. The velocity and calibration energy were fixed at 3.5 m/s and 1 J, respectively.

2.4.6. Melt flow rate measurement (MFR)

The melt flow rate was measured using a capillary rheometer (SRZ-400 C). Before MFR measurements, all the samples were dried in a vacuum oven at 80°C for

6 hours, and then, tested at 255°C with an overhead weight of 2.16 kg. Approximately 8 g of samples was loaded into the equipment at every turn.

3. Results and discussion

3.1. Grafting of POE and SBS

Both in the case of POE and SBS, direct evidence of the grafting was obtained using FT-IR spectra (Figure 2). In comparison with FT-IR spectra of pure POE and SBS, the new absorption band at 1730 cm⁻¹ can be attributed to the carbonyl stretching of the glycidyl methacrylate (GMA) molecule, demonstrating the occurrence of grafting reaction. For SBS-g-GMA, the peak at 1375 cm⁻¹ was not observed, which was an intense absorption of the symmetric -CH₃ stretching in the original SBS sample.

The grafting degree (G_d) was measured by the previously described method and were 3.8% and 2.7% for POE-g-GMA and SBS-g-GMA, respectively. Notably, there were deviations between the test values and real values, which is because in the back-titration method, the result of discoloration is observed via visual observation.

3.2. PET/HDPE blends with and without compatibilizers

3.2.1. Morphology

SEM micrograph of Figure 3 shows the morphology of PET/HDPE blend, wherein, the typical morphology of an immiscible system is observed. HDPE was dispersed as an island phase in the PET matrix. The size of most HDPE particles was irregular and the dispersion in the matrix was not uniform. The particles and cavities on the cryofractured surface were smooth, indicating that the adhesion between the matrix and the dispersed phase was poor.

The morphology of PET/HDPE blends with different compatibilizers content was quite different from that of PET/HDPE blends. Figures 4 and 6 show the SEM micrographs of PET/HDPE/POE-g-GMA and PET/HDPE/SBS-g-GMA blends, respectively. As shown in Figure 4a, with the addition of 5% POE-g-GMA, the diameters of dispersed HDPE particles decreased sharply and the dispersion of HDPE phase was greatly improved, although there were still several smooth cavities. When the POE-g-GMA content was up to 10%, increasingly more black holes were observed around HDPE particles, suggesting that the HDPE particles were encapsulated by POE-g-GMA to form a core-shell structure (Figure 5). The addition of additives not only reduced the size of the dispersed phase but also increased the adhesion between the PET matrix

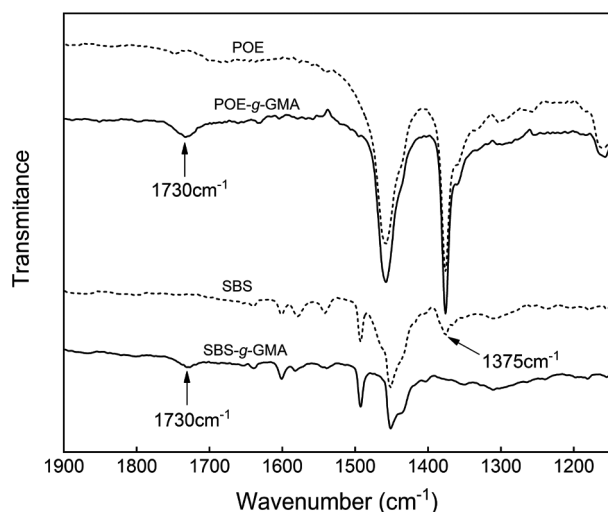


Figure 2. FT-IR spectra of pure POE, POE-g-GMA, pure SBS, and SBS-g-GMA.

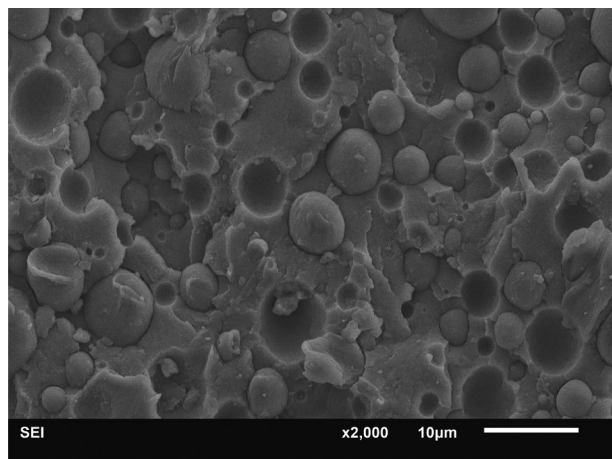


Figure 3. SEM micrograph of PET/HDPE blend.

and HDPE. Several novel fibrillar structures spanning the voids between two phases can be observed in Figure 6, suggesting efficient bonding between the phases. These fibrils were most likely the grafted chains of PET-g-POE-g-GMA produced by the reaction of PET and POE-g-GMA, which is conducive for the reduction of surface tension, providing sterical stabilization and depressing the coalescence.^[31] With increase in POE-g-GMA content, increasingly more PET particles entered into the dispersed HDPE phase, and then, formed salami structure.^[32] Figure 4c shows that the blends contained 15% POE-g-GMA, diameters of HDPE particles decreased continuously, and dispersion of dispersed phase was clearly improved. Numerous HDPE particles encapsulated by POE-g-GMA were uniformly distributed in the PET matrix. Also, large dispersed particles were found, and the voids around these large particles were also wider. As

shown in Figure 4d, with 20% content of compatibilizer, there were several circular particles with different sizes and multiple core-shell particles with large size and tensile shape in the PET matrix. In this case, the compatibilizer was more inclined to self-aggregate rather than encapsulate HDPE particles, leading to a large number of rubber particles in the PET matrix. The reason for this morphology may be the coalescence due to low viscosity of POE-g-GMA.^[33] At the same time, HDPE particles tended to aggregate due to the lack of compatibilizers; thus, forming large particles.

Regarding the morphology of PET/HDPE/SBS-g-GMA blends, which are similar to PET/HDPE/POE-g-GMA blends, with the addition of SBS-g-GMA, the size of dispersed particles decreased and the dispersion of minor phase was improved (see Figure 7a–c). Similar core-shell and fibril-like structures were observed again. As GMA could react with the hydroxyl and carboxyl end groups of PET, grafted chains of PET-g-SBS-g-GMA were also produced in PET/HDPE/SBS-g-GMA blends. This structure is particularly evident in Figure 7c. A similar morphology to Figure 4d also exists in Figure 7d. HDPE particles tended to aggregate due to the lack of compatibilizer and the size of HDPE particles was increased again. The average diameter of dispersed phase particles is presented in Table 3. However, the morphology of the PET/HDPE/SBS-g-GMA blends was quite different from that of the PET/HDPE/POE-g-GMA blends. The diameters of dispersed HDPE particles were larger and more rounded. Also, smooth holes could still be observed. These results suggest that although the addition of SBS-g-GMA reduced the interfacial tension and increased the bonding between the phases, the degree of bonding was not as high as the PET/HDPE/POE-g-GMA blends. The morphological differences led to the differences in mechanical properties of blends compatibilized with two different compatibilizers.

3.2.2. Thermal analysis

As is well-known, PET is a semi-crystalline polymer. The crystallization of PET was found to be affected by the addition of HDPE, POE-g-GMA, and SBS-g-GMA, and consequently, the mechanical properties were affected. The thermal properties of pure PET and its blends were characterized by DSC thermogram measurement. Figure 8a,b demonstrate the non-isothermal crystallization process of PET/HDPE/POE-g-GMA and PET/HDPE/SBS-g-GMA blends, respectively, and the relevant data are summarized in Tables 4 and 5, respectively. The crystallization temperature (T_c) of PET in PET/HDPE blends was 203.3°C, which is about 30°C higher than that for PET homopolymer ($T_c = 172.4^\circ\text{C}$). Moreover, the values of T_c of PET phase in PET/HDPE/

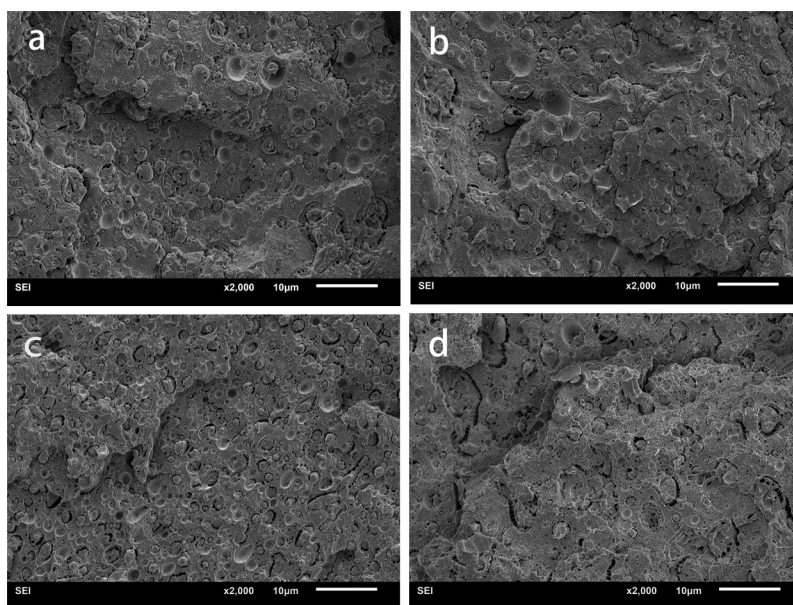


Figure 4. SEM micrographs of PET/HDPE/POE-*g*-GMA blends: (a) 80/20/5, (b) 80/20/10, (c) 80/20/15, (d) 80/20/20. All the above blends were etched by xylene for 48 h.

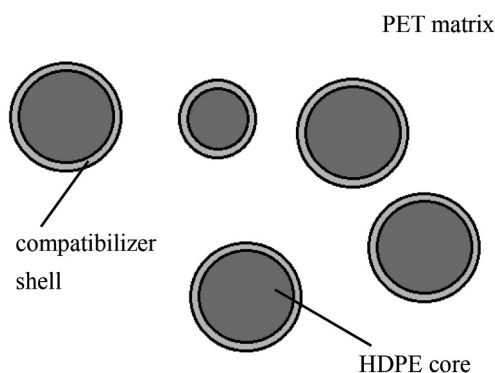


Figure 5. Core-shell structure in PET/HDPE compatibilized blends.

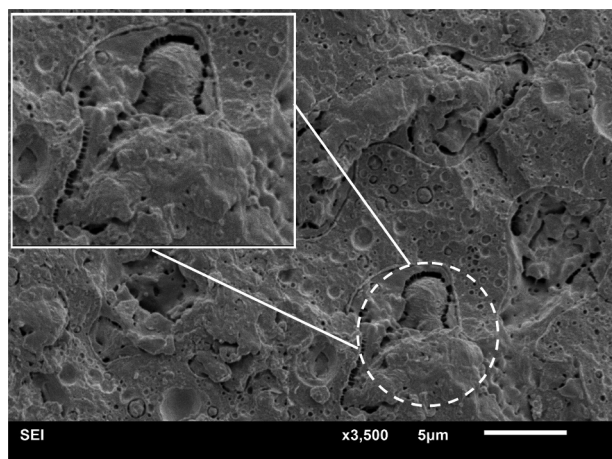


Figure 6. Schematic of fibrillar structure in PET/HDPE/POE-*g*-GMA blends.

POE-*g*-GMA and PET/HDPE/SBS-*g*-GMA blends were 20–30°C higher than that of PET homopolymer. These results demonstrate that the introduction of additives allowed the crystallization to occur earlier as compared to the PET homopolymer. The increase in T_c of PET phase in blends can be ascribed to the dispersed particles playing the role of heterogeneous nucleating agents in PET matrix. The values of T_c of PET phase in PET/HDPE/POE-*g*-GMA blends were lower than T_c of PET phase in PET/HDPE blends ($T_c = 203.3^\circ\text{C}$). However, for PET/HDPE/SBS-*g*-GMA blends, with the addition of SBS-*g*-GMA, the values of T_c of PET phase continued to increase. They began to be lower than those in PET/HDPE blends after the content of SBS-*g*-GMA reached 15%.

As presented in Tables 4 and 5, the values of crystallization enthalpy (ΔH_c) and crystallinity of PET phase in PET/HDPE/POE-*g*-GMA blends decreased with the addition of POE-*g*-GMA. This is probably because the grafted chains of PET-*g*-POE-*g*-GMA were formed by the reaction of PET matrix and GMA molecules, leading to a decrease in the regularity of the PET molecular chains; thereby, hindering the process of crystallization. The values of ΔH_c and crystallinity of PET phase in PET/HDPE/SBS-*g*-GMA blends continued to increase before the content of SBS-*g*-GMA reached 15%. These results suggest that the grafted chains of PET-*g*-SBS-*g*-GMA were not as effective as the grafted chains of PET-*g*-POE-*g*-GMA in inhibiting the orientate alignment of the PET molecular chains. At the same time, the reaction between the PET matrix and the GMA molecule induced

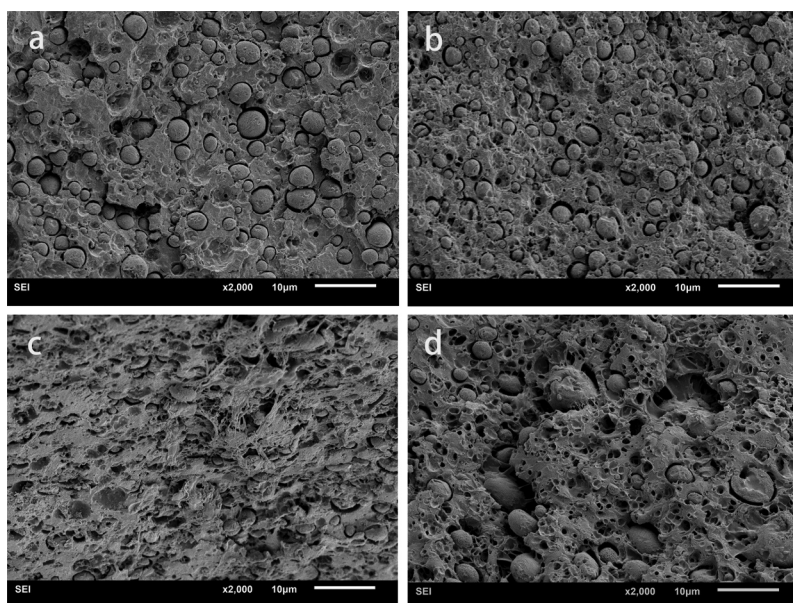


Figure 7. SEM micrographs of PET/HDPE/SBS-*g*-GMA blends: (a) 80/20/5, (b) 80/20/10, (c) 80/20/15, (d) 80/20/20. All the above blends were etched by xylene for 48 h.

Table 3. The average diameter of dispersed phase particles.

Samples	Blend composition	Average diameter (μm)
PET/HDPE	80/20	5.4 ± 1.3
PET/HDPE/POE- <i>g</i> -GMA	80/20/5	2.2 ± 0.8
	80/20/10	1.9 ± 0.7
	80/20/15	1.1 ± 0.2
	80/20/20	3.6 ± 1.6
PET/HDPE/SBS- <i>g</i> -GMA	80/20/5	3.0 ± 1.2
	80/20/10	2.3 ± 0.8
	80/20/15	1.7 ± 0.6
	80/20/20	4.0 ± 1.2

more PET dispersed in the SBS-*g*-GMA phase to further increase the activity of the PET molecular chains. The T_m values of PET in PET/HDPE/POE-*g*-GMA were lower than those in PET/HDPE/SBS-*g*-GMA, which can be attributed to the differences in crystallinity of PET in two types of blends.

As HDPE is also a semi-crystalline polymer, compatibilizers not only affect the crystallization of PET phase in blends but also affect the crystallization of HDPE. The thermal properties of HDPE phase for PET/HDPE/POE-*g*-GMA and PET/HDPE/SBS-*g*-GMA blends are summarized in Tables 6 and 7, respectively. The values of T_m indicate that the type and the amount of compatibilizer had no significant effect on the melting point of HDPE phase in the blends. Addition of both types of compatibilizers reduced the crystallinity of HDPE phase in the blends. On one hand, the rigidity of PET molecular segments was higher than that of HDPE, which hindered the orderly arrangement of HDPE molecular segments. On the other hand, POE has

a PE molecular chain structure like HDPE, and other polymer segments which were connected to the PE segment in the POE inevitably hindered the crystallization process of the HDPE phase. Therefore, the crystallinity of HDPE phase in PET/HDPE/POE-*g*-GMA blends was more visibly decreased. However, the crystallinity of HDPE phase in the blends increased gradually with the increase in content of compatibilizers. During the cooling process, the formed PET grains acted as nucleating agent for HDPE; thereby, promoting the crystallization of HDPE. Moreover, polar PET had an electrostatic effect on the HDPE molecular chains, which led to the aggregation, alignment, and orientation of HDPE molecular chains, all of which entered the compatibilizer shell around the PET crystal (Figure 9). Therefore, more is the content of compatibilizer, thicker is the compatibilizer shell, and more HDPE molecular chains enter the compatibilizer shell. Thus, the crystallinity of the HDPE phase increased with the increase in compatibilizer amount.

Figure 10a,b show the second heating progress of PET and both types of blends. There was a single melting peak for PET homopolymer. Two melting peaks for PET could be found in most blends. This is perhaps due to partial melting and recrystallization of PET crystallite during the heating process.^[34] However, when the content of POE-*g*-GMA exceeded 15%, the double melting peaks for PET in PET/HDPE/POE-*g*-GMA blends gradually integrated and formed a novel melting peak. Nevertheless, no similar phenomenon occurred in the PET/HDPE/SBS-*g*-GMA

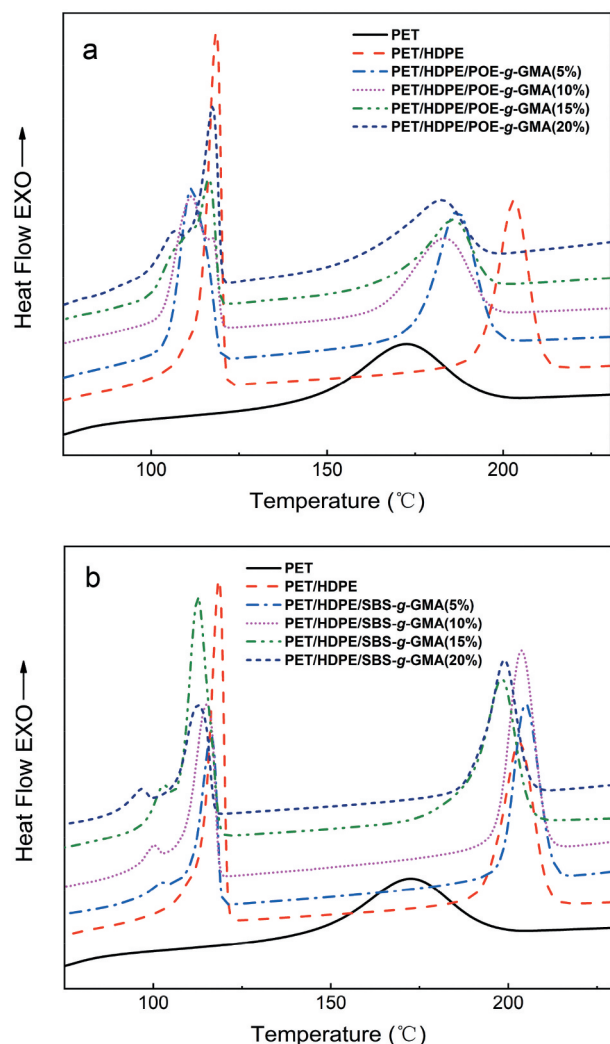


Figure 8. DSC cooling thermograms of PET and blends. (a) PET/HDPE/POE-g-GMA, (b) PET/HDPE/SBS-g-GMA.

blends. Therefore, as per the DSC test results, POE-g-GMA had a more clear effect on the compatibilization of PET/HDPE blends.

3.2.3. Mechanical properties

The effects of POE-g-GMA and SBS-g-GMA contents on the mechanical properties of PET/HDPE blends are discussed in detail. Figure 11 shows the notched Izod impact strength with or without compatibilizers.

Table 5. Thermal properties data of PET in PET/HDPE/SBS-g-GMA blends.

Blend composition	PET				
	T_c (°C)	ΔH_c (J/g)	ΔH_m (J/g)	T_m (°C)	Crystallinity (%)
80/20/5	204.8	30.7	−29.6	243.3, 250.9	27.8
80/20/10	203.7	35.7	−33.5	242.8, 249.8	33.1
80/20/15	198.0	35.8	−31.9	241.2, 249.0	32.8
80/20/20	198.8	26.7	−25.2	239.2, 247.4	27.0

Table 6. Thermal properties data of HDPE in PET/HDPE/POE-g-GMA blends.

Blend composition	HDPE		
	T_m (°C)	ΔH_m (J/g)	Crystallinity (%)
80/20	132.5	−39.8	69.0
80/20/5	129.5	−17.2	33.8
80/20/10	129.1	−19.5	36.5
80/20/15	132.5	−22.8	44.7
80/20/20	132.5	−22.7	46.4

Table 7. Thermal properties data of HDPE in PET/HDPE/SBS-g-GMA blends.

Blend composition	HDPE		
	T_m (°C)	ΔH_m (J/g)	Crystallinity (%)
80/20/5	131.5	−23.9	42.9
80/20/10	129.8	−24.9	46.7
80/20/15	129.0	−29.6	58.1
80/20/20	130.1	−28.8	59.0

As shown in Figure 11, as the compatibilizers content increased from 5% to 20%, the notched Izod impact strength of the blends first increased and then decreased. The notched Izod impact strength of PET/HDPE blend was extremely low at 19.3 J/m. When the content of POE-g-GMA was 15%, the notched Izod impact strength of PET/HDPE/POE-g-GMA blends increased significantly to 110.8 J/m. This is because the graft chains of PET-g-POE-GMA produced by the reaction between PET and POE-g-GMA could reduce the interfacial tension and improve the dispersion of the dispersed phase. As shown in Figure 4c, numerous HDPE particles encapsulated by POE-g-GMA were evenly dispersed in the PET matrix. These core-shell particles played a critical role in transferring the applied stress and inhibiting the formation of harmful cracks. However,

Table 4. Thermal properties data of pure PET and PET in PET/HDPE/POE-g-GMA blends.

Blend composition	PET					
	T_c (°C)	ΔH_c (J/g)	ΔH_m (J/g)	T_m (°C)	T_{sm}^a (°C)	Crystallinity ^b (%)
pure PET	172.4	32.8	−44.0	247.2	−	31.4
80/20	203.3	35.0	−31.1	244.3, 246.2	−	27.8
80/20/5	186.7	35.0	−34.7	239.3, 246.2	165.6	32.5
80/20/10	183.4	28.0	−27.9	237.1, 243.4	165.5	27.4
80/20/15	185.9	23.1	−23.1	243.7	167.1	23.7
80/20/20	182.3	23.8	−24.2	243.3	167.3	25.9

T_{sm}^a : Melting point of secondary crystallization of PET

Crystallinity^b of secondary crystallization of PET is not involved.

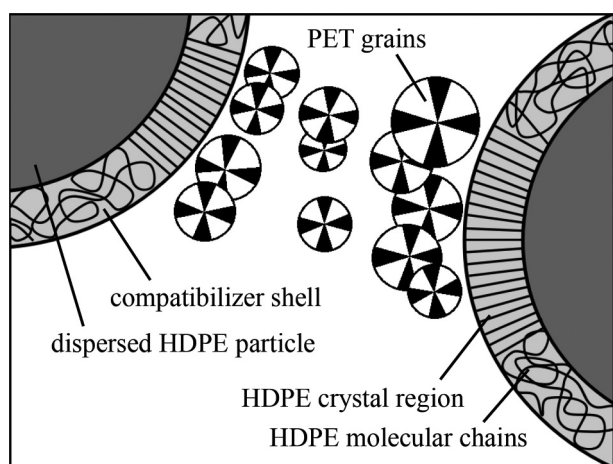


Figure 9. Schematic of HDPE crystallization behavior in compatibilizer shell.

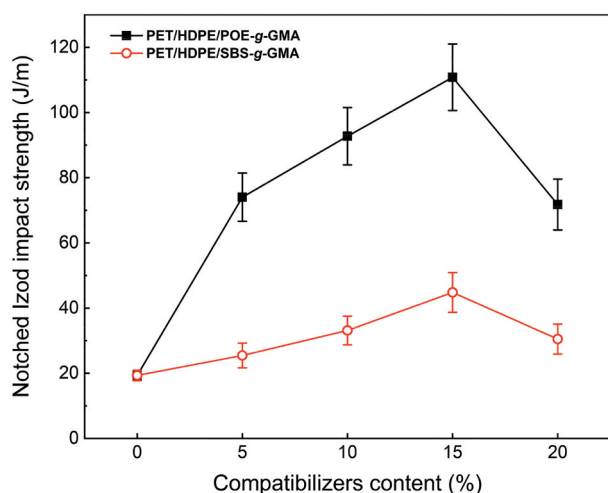


Figure 11. The notched Izod impact strength of PET/HDPE blends with and without compatibilizers.

the notched Izod impact strength of PET/HDPE/SBS-g-GMA blends only slightly increased, and the highest notched Izod impact strength was 44.8 J/m at the SBS-g-GMA content of 15%. This phenomenon may be due to the lower miscibility of the SBS with the HDPE and lower grafting degree of SBS-g-GMA in comparison to POE-g-GMA. When the two compatibilizers were over-used (20%), a large number of rubber particles were irregularly distributed in the PET matrix. Under an external loading, rubber particles caused stress concentration, resulting in a large number of crazes. These large-scale core-shell particles in Figures 4d and 7d could not effectively prevent the development of crazes into deleterious cracks, resulting in degradation of impact properties of blends. The difference in notched Izod impact strength between the two types of blends is

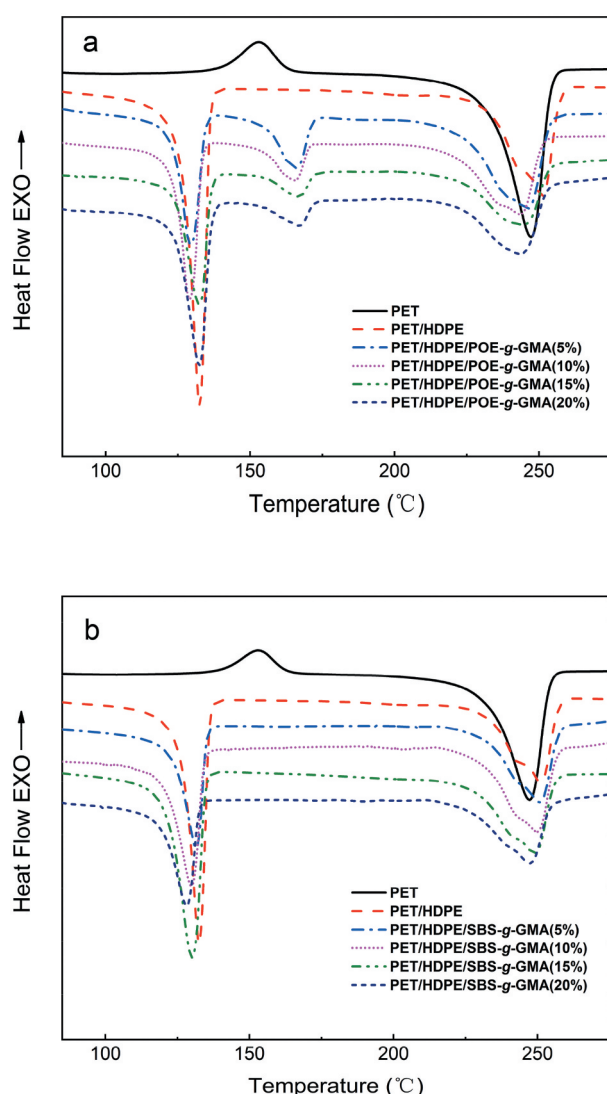


Figure 10. The second DSC heating thermograms of PET and blends (a) PET/HDPE/POE-g-GMA, (b) PET/HDPE/SBS-g-GMA.

consistent with the phase morphology exhibited by their SEM micrographs.

The effects of POE-g-GMA and SBS-g-GMA contents on the tensile properties of PET/HDPE blends were also investigated in depth. As shown in Table 8, with the increase in compatibilizers content, the transformation of elongation at break of blends was similar to that of the notched Izod impact strength of blends. The elongation at break of uncompatibilized PET/HDPE blend was only 5.8% because of the poor adhesion between PET and HDPE. While the addition of POE-g-GMA and SBS-g-GMA could effectively increase the elongation at break of PET/HDPE blends. When 15% POE-g-GMA or 15% SBS-g-GMA was added in PET/HDPE blends, the elongation at break increased from 5.8% to 41.4% and 36.3%, respectively. With the addition of compatibilizers, the tensile strength of both PET/HDPE/POE-

Table 8. Tensile properties of PET/HDPE blends with and without compatibilizers.

Samples	Blend composition	Tensile strength (MPa)	Elongation at break (%)
PET/HDPE	80/20	40.5 ± 1	5.8 ± 0.2
PET/HDPE/POE-g-GMA	80/20/5	40.6 ± 0.3	34.9 ± 5.0
	80/20/10	37.6 ± 0.5	36.2 ± 3.9
	80/20/15	34.8 ± 0.1	41.4 ± 2.8
	80/20/20	29.4 ± 0.8	13.8 ± 2.2
PET/HDPE/SBS-g-GMA	80/20/5	40.8 ± 0.3	6.5 ± 0.6
	80/20/10	35.1 ± 0.7	24.1 ± 3.9
	80/20/15	30.1 ± 0.2	36.3 ± 6.3
	80/20/20	27.1 ± 0.2	16.6 ± 3.0

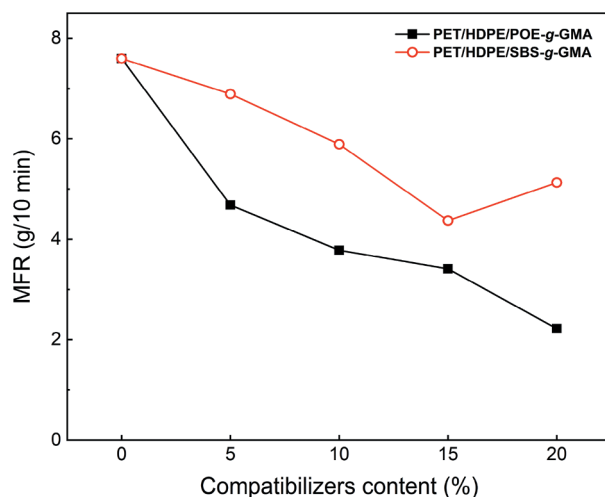
g-GMA and HDPE/HDPE/SBS-g-GMA blends decreased. As both POE and SBS are elastomers, higher the content of POE-g-GMA and SBS-g-GMA in PET/HDPE blend, lower is the tensile strength of the blend. Also, the toughness and elongation at break of PET/HDPE blends were clearly improved by the addition of POE-g-GMA or SBS-g-GMA, with a decrease in the tensile strength.

3.2.4. Melt flow rate

The melt flow rate values can reflect on the compatibility of the blends. Figure 12 shows the effect of compatibilizers content on the melt flow rate of blends. Although both POE-g-GMA and SBS-g-GMA reduced the melt flow rate of blends, the effect of POE-g-GMA was more pronounced. Owing to the presence of graft copolymer produced by the reaction of the epoxy groups of compatibilizers and PET matrix at the interface of PET and HDPE, the relative molecular mass of the blends increased; therefore, the melt flow rate was reduced. As shown in Figure 12, for PET/HDPE/POE-g-GMA blends, with the increase in POE-g-GMA content, the values of MFR decreased rapidly from 7.6 g/10 min to 2.2 g/10 min. In contrast, the MFR values of PET/HDPE/SBS-g-GMA gradually decreased from 7.6 g/10 min to 4.4 g/10 min, and then, rose slightly from 4.4 g/10 min to 5.1 g/10 min. Therefore, the melt flow rate test results further confirm that POE-g-GMA was better than SBS-g-GMA in compatibilizing the PET/HDPE blends.

4. Conclusion

Herein, both POE-g-GMA and SBS-g-GMA have been used as reactive compatibilizers for the PET/HDPE blends. With the addition of compatibilizers, the morphology of blends evolved from the island morphology to core-shell and fibril-like morphology with HDPE particles encapsulated by both POE-g-GMA and SBS-g-GMA. The

**Figure 12.** Effect of compatibilizers content on MFR of PET/HDPE blends.

appearance of fibrous structure implies the formation of graft copolymers of PET-g-POE-g-GMA or PET-g-SBS-g-GMA, which reduced the interfacial tension and increased the adhesion between the PET matrix and HDPE particles. Notably, excessive compatibilizers can induce minor phase agglomerates to form larger domains. DSC analysis reveals that HDPE particles can act as a nucleating agent to improve the crystallization temperature of PET matrix. The addition of POE-g-GMA effectively hindered the crystallization of PET phase, while the addition of SBS-g-GMA had a relatively little influence on the crystallization behavior of PET phase in PET/HDPE blends. Regarding the mechanical properties, the toughness of PET/HDPE blend was remarkably improved upon the addition of selected compatibilizer, especially, the notched Izod impact strength, while the tensile strength was reduced. Compared to PET/HDPE blends, addition of POE-g-GMA (15%) to the blend led to best mechanical properties with impact strength and elongation at break of 110.8 J/m and 41.4%, respectively. The values of MFR indicate that both POE-g-GMA and SBS-g-GMA enhanced the melt viscosity of PET/HDPE blends, with POE-g-GMA having a more pronounced effect. Overall, the results presented here can serve as a platform for using POE-g-GMA and SBS-g-GMA as reactive compatibilizers for PET/HDPE blends. Both POE-g-GMA and SBS-g-GMA ensured anchorage of the dispersed phase in PET/HDPE immiscible blends.

Funding

This work was supported by Jilin Scientific and Technological Development Program under Grant No. 20180201075SF and

Jilin Province Development and Reform Commission under Grant No. 2019C043-8.

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